Faculdade de Design,
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01 - Introduction to Machine Learning

Automatic Applied Learning

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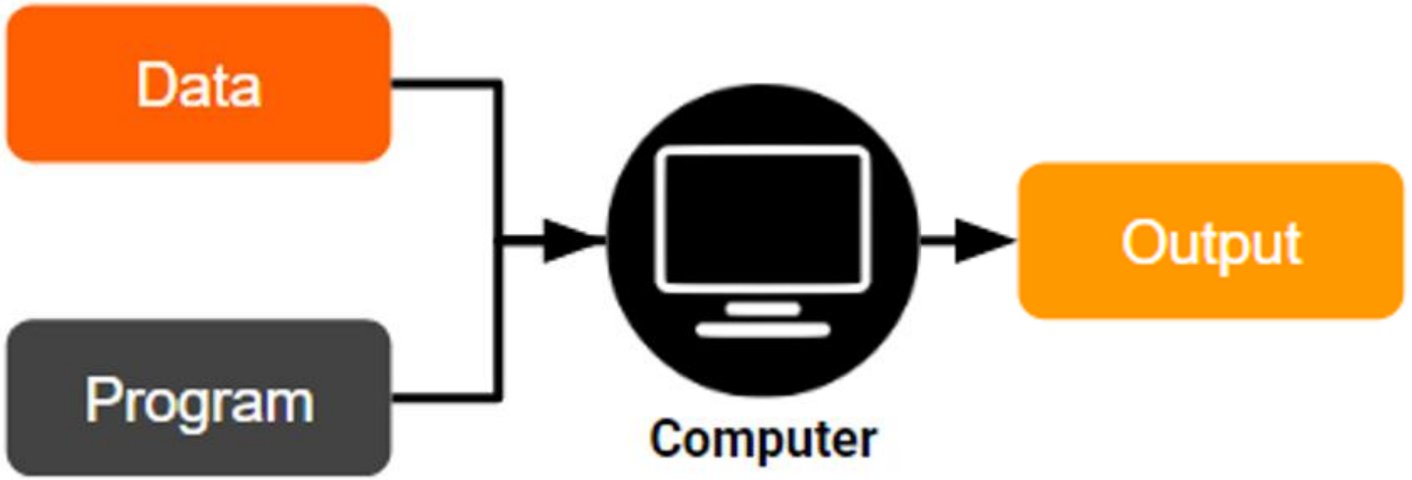
Agenda

- Understand the basic idea of machine learning
- Compare traditional programming with machine learning
- Distinguish between:
  - Supervised learning
  - Unsupervised learning
- Understand common machine learning terms
- Identify examples and real-world applications
- Recognize both the possibilities and limitations of machine learning

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How works a computer program?

- In traditional programming, a person writes **explicit instructions for the computer**.

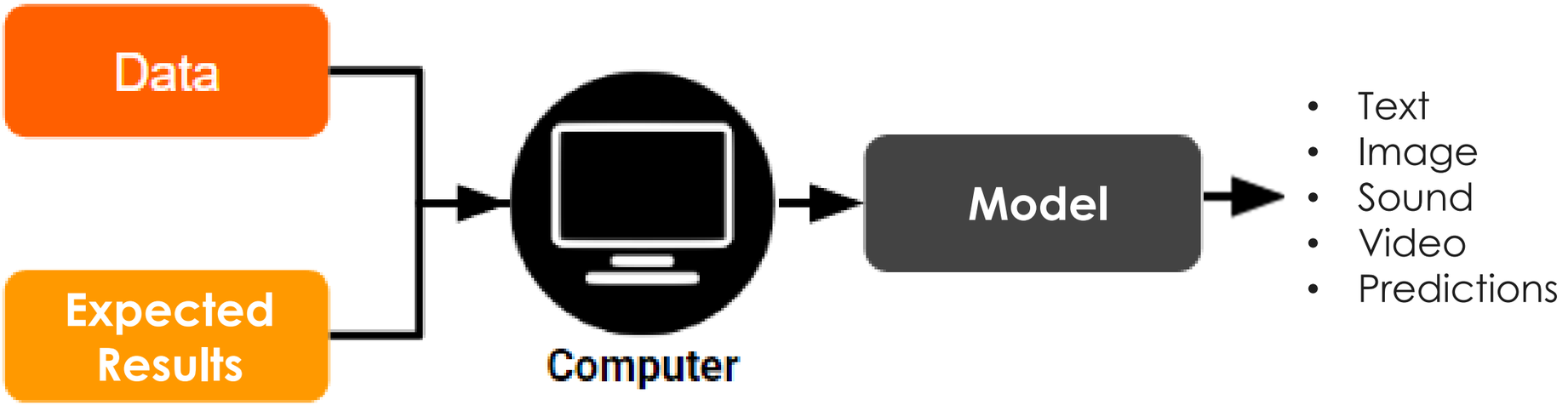


- Text
- Image
- Sound
- Video
- Predictions

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What is Machine Learning?

- In machine learning, the computer receives **data** and **examples of expected results**.



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What is Machine Learning?

- **Machine learning** (ML) trains **software models** to **make predictions** or **generate content from data**.

- **Unlike traditional programming**, which requires explicitly modeling complex rules - such as **atmospheric physics to predict rainfall - ML learns patterns** from **large datasets**.

- **Once trained**, **a model** can use **current data** to produce **something**, like **predictions** to **estimate the amount of rain**.

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Main difference

- **Traditional programming**
  - Humans write the rules.

- **Machine learning**
  - The computer learns rules from examples.

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An Analogy

- How to teach to a child what is an apple?
- We can specify **rules to recognize it**:
  - It's normally round
  - It may be red, green, or yellow
  - It is often smaller than a watermelon
  - However, real objects do not always follow simple rules.



Red plum



Yellow plum



Red nectarine

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An Analogy

- How to teach to a child what is an apple?
- **Another approach** is to show the child **many examples:**

"This is an apple."
"These are also apples."



"These are not apples."



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Again, What Is Machine Learning?

- **Machine learning** is a way of creating **computer systems that learn patterns from data** and use **those patterns to make predictions**, **decisions**, or **generate results**.

- A machine learning system is not simply **memorizing instructions**. **It uses examples** to **build a model**.

- The **generated model can then be applied to classify new data** that it has not seen before.

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What Does “Learning” Mean?

In machine learning, **learning** means:

- **Receiving data**
- Finding **patterns** or **relationships**
- **Building** a model
- **Evaluating** the model
- **Using** the model with **new data**
- **Improving** the model when necessary
- The system **does not learn in the same way** as a human:
  - it calculates relationships based on data and mathematical methods.

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Important Terms

Data

- Information used by a machine learning system.
  - Examples:
    - Images
    - Text
    - Audio
    - Prices
    - Medical measurements
    - Customer transactions

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Important Terms

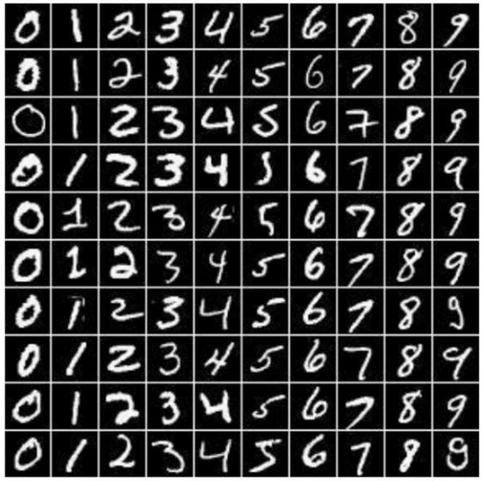
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Dataset

- A **collection of data** used for **analysis or model development**.

MNIST dataset

The MNIST database of handwritten digits,
available from this page,
has a training set of 60,000 examples, and a test set of 10,000 examples.
It is a subset of a larger set available from NIST.
The digits have been size-normalized and centered in a fixed-size image.



Cats dataset



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Features and Labels

Feature

- A **measurable characteristic used** by the model.
- Examples for predicting house prices:
  - Number of bedrooms
  - Property size
  - Location
  - Age of the Building

Label

- The correct answer associated with an example.
- For house-price prediction: the **actual sale price is the label**.

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Model

- A model is the **result of the learning process**.
- It represents **patterns found in the data** and **can be used to make predictions** or **decisions**.

Example

- A **model trained with information about houses** may **learn how factors** such as **size**, **location**, and **number of bedrooms relate to price.**

- The model does not **know the house in a human sense**. **It uses patterns found** in previous examples.

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Training and Inference

Training

- The model **learns from examples** in a dataset.

Training data → Learning algorithm → Trained model

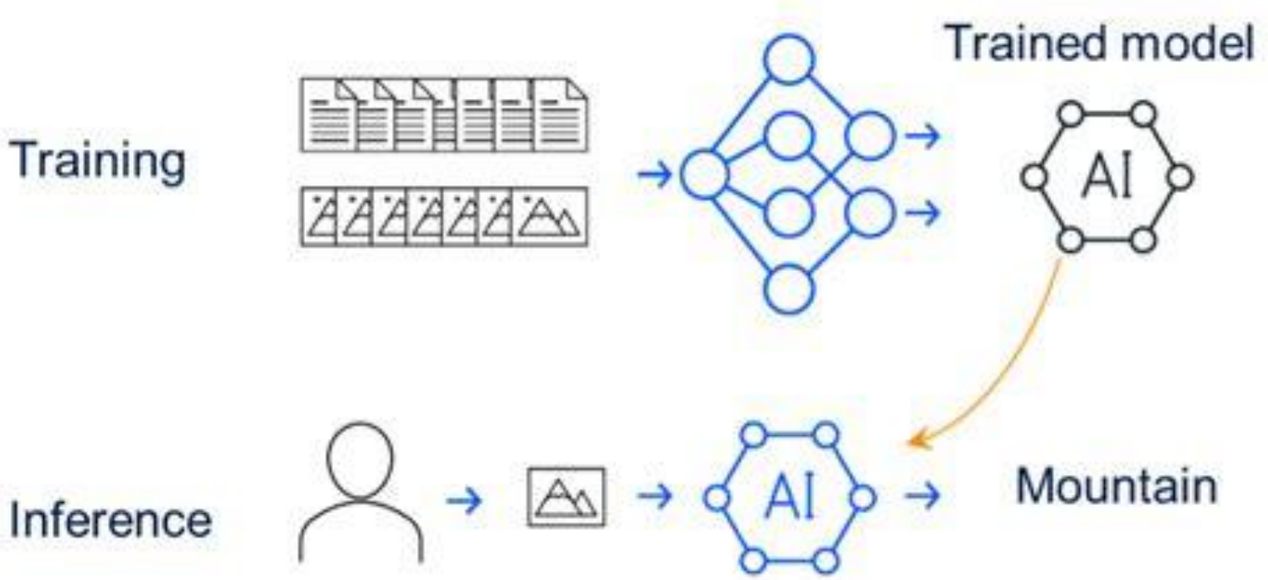
Inference or **prediction**

- The **trained model is used with new data**.

New data + Trained model → Prediction

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Training and Inference



- **Training**: Learn from thousands of labeled house sales
- **Inference/prediction**: Estimate the price of a new house

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Algorithm and Model

- These terms are **related but different**.

Algorithm

- A **method** or **procedure** used **to learn from data**.
  - Decision tree
  - Linear regression
  - K-means clustering
  - Neural network

Model

- The **result produced** after applying an algorithm to data.

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Algorithm and Model

- **Algorithm**: The learning method

- **Model**: What the method learned from the data

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Supervised Learning

- In **supervised learning**, the **training data** includes the **correct answers**.

Features + Labels → Learning algorithm → Model

- The model **learns the relationship** between **inputs** and **outputs**.

Size	Location	Actual price
80 m²	Big city	€ 450.000
120 m²	Small city	€ 250.000
60 m²	Big city	€ 350.000

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Supervised Learning Tasks

Classification

- The model predicts a **category** or **class**.
  - Spam or not spam
  - Disease or no disease
  - Cat, dog, or bird
  - Approved or rejected loan

Regression

- The model predicts a numerical value.
  - House price
  - Temperature
  - Sales revenue
  - Delivery time

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Classification Example: Email Spam

Training data

- The system receives emails **labeled** as:
  - **Spam**
  - **Not spam**

- It may examine **features** such as:
  - Words in the message
  - Number of links
  - Sender information
  - Message structure
  - Previous user actions

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Classification Example: Email Spam

Prediction

- For a **new email**, the model **predicts**:



Not SPAM



SPAM

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Regression Example: House Prices

A **regression model predicts a number**.

- **Input features**
  - Property size
  - Number of bedrooms
  - Location
  - Parking spaces
  - Age of the building

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Regression Example: House Prices

A **regression model predicts a number**.

- **Input features**
  - Property size
  - Number of bedrooms
  - Location
  - Parking spaces
  - Age of the building
- **Output**
  - Estimated price: $325,000

- The prediction **may not be precise**. It is an estimate based on patterns in previous examples.

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Unsupervised Learning

- In unsupervised learning, the data **does not include correct answers** or **labels**.

Data → **Learning algorithm** → **Patterns** or **groups**

- The system **searches** for **structure in the data**.

Example

- A store may have **customer information** but **no predefined customer categories**.
- An algorithm **can group customers according to similarities** in their behavior.

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Clustering

- **Clustering** is an **unsupervised learning technique** that groups **similar examples**.

Example: Customer groups

A business may identify groups such as:

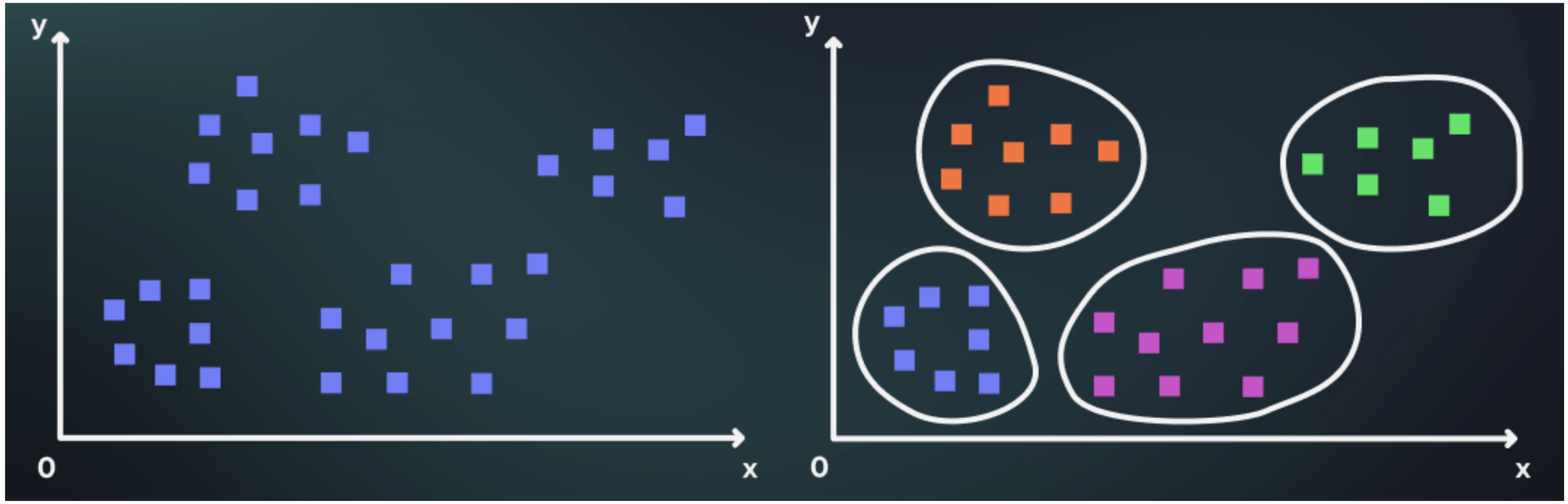
- Customers who buy frequently
- Customers who spend a lot
- Customers who purchase only during promotions
- New customers with limited activity

The algorithm **finds these groups based on the available data**.

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Clustering

- **Clustering** is an **unsupervised learning technique** that groups **similar examples**.



The algorithm **finds these groups based on the available data**.

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Other Unsupervised Learning Uses

Unsupervised learning can help with:

- **Customer segmentation**
- **Document organization**
- **Image grouping**
- **Detecting unusual behavior**
- **Finding hidden patterns**
- **Reducing the number of variables in a dataset**

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Supervised vs. Unsupervised Learning

Question	Supervised Learning	Unsupervised Learning
Are labels available?	**Yes**	**No**
Main goal	Predict an answer	Find patterns or groups
Common tasks	Classification and regression	Clustering and pattern discovery
Example	Predict whether a message is spam	Group similar messages
Evaluation	Compare predictions with known answers	Analyze the quality and usefulness of discovered patterns

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